Average Adventure

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Momen-G-Ar's little brother asks for help with his math homework. The homework provides an array A of N integers and requires the following:

- split the elements of the array into three categories:
  - **ABOVE:** elements that are greater than the average of the array.
  - **BELOW:** elements that are less than the average of the array.
  - **EQUAL:** elements that are equal to the average of the array.
- Count how many numbers are in each category.

However!! Momen is currently working on an **urgent** task for the company he is working for. so he asks you to help his brother doing his homework.

Note: if any floating points appears after calculating the average you should ignore them (41.3 becomes 41) then print your answer according to that.

Input

Input consisting of two lines:

The first line contains single integer N $(1 \leq N \leq 10^5)$ the number of elements in the array.

The second line contains N values $A_1, A_2, ...A_N$ $(-10^5 \leq A_I \leq 10^5)$ the array elements.

Output

Print **two lines** per category:

- First line: number of elements in the category followed by the category name.
- Second line: the elements in that category, space-separated.

You have to print the categories in the following order: `ABOVE`, `BELOW`, `EQUAL`.

Example

standard input	standard output
5 2 5 7 3 9	2 ABOVE 7 9 2 BELOW 2 3 1 EQUAL 5

Note

Consider this array [2, 5, 7, 3, 9]:

- The average is: 5.2
- Average without the decimal point: 5

- Numbers above the average: 7, 9
- Numbers below the average: 2, 3
- Numbers equal to the average: 5